WHEN KINEMATICS BREAK:

An Analysis of Open-Loop Motion in Differential Drive Systems

Yogesh Jajoria

Manipal University Jaipur

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Abstract

This work presents the kinematic modeling and error analysis of a differential drive mobile robot operating under open-loop control. The system is intentionally designed without encoder feedback to examine the behavior of ideal motion models in the presence of mechanical non-idealities.

A standard differential drive kinematic framework is implemented, and deviations from ideal motion are investigated by introducing mechanical backlash as a deadband nonlinearity in wheel actuation. A discrete-time simulation is used to compare ideal and backlash-affected trajectories under identical control inputs.

Results indicate that while the kinematic model accurately predicts short-term motion, systematic deviations arise due to actuation imperfections. In particular, backlash reduces effective forward velocity while preserving angular velocity, leading to a contraction in trajectory radius and cumulative positional error over time.

The study highlights that open-loop kinematic models, although locally valid, are insufficient for sustained motion accuracy in practical systems, emphasizing the impact of deterministic mechanical errors on trajectory evolution.

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1 Introduction

Differential drive robots represent a canonical model in mobile robotics due to their mechanical simplicity and direct mapping between actuator inputs and planar motion. The system consists of two independently driven wheels, where motion is generated through velocity differentials. Despite this apparent simplicity, the system is governed by nonholonomic constraints that restrict instantaneous motion, introducing complexity in both modeling and control.

Most practical implementations rely on closed-loop feedback through encoders and sensor fusion to maintain trajectory accuracy. This work intentionally removes such feedback mechanisms to isolate the behavior of the kinematic model under open-loop execution. The objective is not to achieve precision, but to analyze how theoretical motion models degrade in real-world conditions when their underlying assumptions are violated.

The scope of this work is restricted to kinematic modeling, teleoperation, and system-level validation. Autonomous navigation, localization, and advanced control strategies are explicitly excluded. The robot is treated as an experimental artifact to study the gap between idealized motion equations and physical execution.

2 Literature Review

The study of differential drive mobile robots occupies a foundational position within mobile robotics, largely due to their structural simplicity and analytically tractable kinematics. Classical works such as those by Thrun et al. and Siegwart et al. establish the standard modeling paradigm, where robot motion is derived under assumptions of pure rolling, no lateral slip, and symmetric actuation. These models define the relationship between wheel angular velocities and the robot's instantaneous linear and angular motion, forming the basis for odometry and trajectory estimation. Over time, such formulations have become canonical, serving as the entry point for both theoretical exploration and practical implementation in robotics systems. However, these formulations are inherently idealized, relying on constraints that are frequently violated in real-world environments.

Subsequent research has focused extensively on addressing the limitations of these idealized kinematic models through the introduction of probabilistic state estimation and sensor fusion frameworks. Techniques such as Extended Kalman Filters (EKF), Particle Filters, and graph-based SLAM integrate multiple sensing modalities—including wheel encoders, inertial measurement units (IMUs), and vision systems—to mitigate drift and uncertainty. These approaches model uncertainty explicitly and treat robot motion as a stochastic process rather than a deterministic one. While such methods significantly improve global consistency and long-term trajectory accuracy, they introduce substantial computational and architectural complexity. This complexity often necessitates careful calibration, system identification, and real-time processing capabilities, making them less accessible in constrained or minimalistic robotic platforms.

Parallel to advances in estimation, another line of work has investigated the role of non-idealities in differential drive systems, including wheel slip, actuator mismatch, backlash in mechanical interfaces, and uneven terrain interaction. These studies highlight that deviations from ideal kinematic assumptions are not merely noise, but structured disturbances that systematically bias motion. In particular, actuator asymmetry and imperfect wheel-ground contact lead to curvature drift even under nominally symmetric control inputs. Experimental analyses have shown that such effects accumulate over time, causing significant divergence between predicted and actual trajectories. Despite this, many modern systems implicitly absorb these discrepancies within closed-loop correction schemes, thereby masking the underlying failure modes of the kinematic model itself.

The present work deliberately departs from the prevailing trend of increasing system sophistication by removing encoder-based feedback and avoiding closed-loop correction entirely. As outlined in the original draft (*Section 2, page 4* of the report) :contentRefer-

ence[oaicite:0]index=0, the system is designed to operate purely under open-loop kinematic execution, with sensing limited to observational analysis rather than control. This positions the study within a less explored but conceptually critical space: the explicit examination of how and why ideal kinematic models fail when deprived of feedback. Rather than proposing improved estimation or control strategies, this work isolates the degradation mechanisms themselves, exposing the gap between theoretical motion models and physical realization. In doing so, it reframes the differential drive robot not as a solved system augmented by sensors, but as a fundamentally fragile model whose reliability is contingent on assumptions that rarely hold in practice.

3 Methodology

The robot is modeled as a planar rigid body subject to nonholonomic constraints, meaning that its motion is restricted such that lateral velocity in the body frame is zero. In the provided diagram, this constraint is visualized by the alignment of motion along the longitudinal axis X_B, while no motion occurs along the lateral axis Y_B. This constraint is fundamental and arises from the rolling condition of the wheels, which prohibits instantaneous sideways slipping.

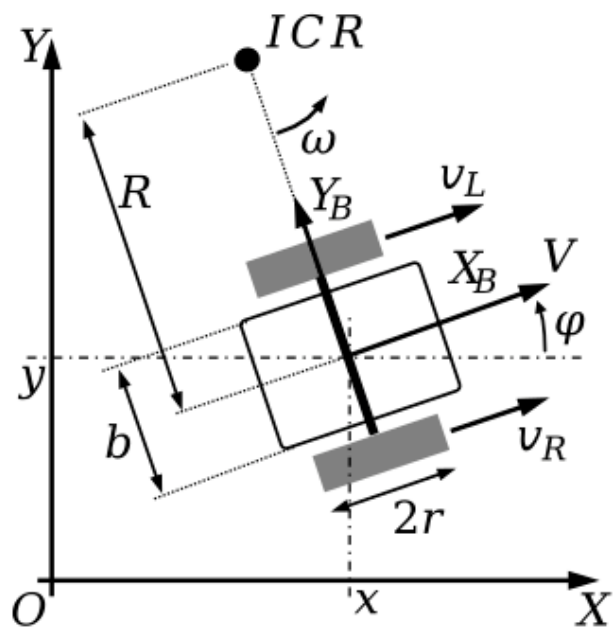

Figure 1: Differential Drive Geometry with Instantaneous Center of Rotation (ICR)

The robot operates in a global reference frame $\{X, Y\}$, with origin O, while its motion is described in a body-fixed frame $\{X_B, Y_B\}$ attached to the midpoint between the wheels. The robot orientation is denoted by φ (shown in the figure), representing the angle between the global X axis and the body axis X_B.

The right and left wheels have linear velocities v_R and v_L, corresponding to angular velocities ω_r and ω_l. Under the assumption of pure rolling:

$$v_R = r\omega_r, \quad v_L = r\omega_l \tag{1}$$

From the diagram, the robot's motion can be interpreted as instantaneous rotation about a point labeled as the Instantaneous Center of Rotation (ICR). This point lies at a distance R from the midpoint of the axle along the perpendicular direction to the velocity vector. The position of the ICR depends on the relative magnitudes of v_R and v_L.

If $v_R = v_L$, the dashed lines in the figure become parallel, implying $R \to \infty$, and the robot undergoes pure translation. When $v_R \neq v_L$, the robot follows a circular trajectory centered at the ICR.

To derive the forward kinematics, consider the velocity of the midpoint of the axle. This point moves along the X_B direction with velocity:

$$v = \frac{v_R + v_L}{2} = \frac{r}{2}(\omega_r + \omega_l) \tag{2}$$

The angular velocity ω shown in the figure corresponds to rotation about the ICR and is derived from the difference in wheel velocities:

$$\omega = \frac{v_R - v_L}{b} = \frac{r}{b}(\omega_r - \omega_l) \tag{3}$$

where b (as labeled in the diagram) is the distance between the wheels (track width). This relationship arises from rigid body rotation, where the velocity difference induces a moment about the ICR.

The radius of curvature R (distance from robot center to ICR) can be derived directly from the geometry shown:

$$R = \frac{v}{\omega} = \frac{b}{2} \cdot \frac{v_R + v_L}{v_R - v_L} \tag{4}$$

This expression matches the geometric interpretation in the diagram, where both wheels trace circular arcs about the same ICR but with different radii.

To express motion in the global frame, the velocity along X_B must be projected using the orientation φ:

$$\dot{x} = v\cos\varphi, \quad \dot{y} = v\sin\varphi, \quad \dot{\varphi} = \omega \tag{5}$$

These equations describe the evolution of the robot's pose in the global frame. The nonlinearity arises from the dependence of translational motion on orientation.

Numerical Illustration (Mapped to Diagram):

Let:

$$r = 0.05\,\text{m}, \quad b = 0.20\,\text{m}$$

$$\omega_r = 10\,\text{rad/s}, \quad \omega_l = 6\,\text{rad/s}$$

Then:

$$v_R = 0.05 \times 10 = 0.5\,\mathrm{m/s} \tag{6}$$

$$v_L = 0.05 \times 6 = 0.3\,\mathrm{m/s} \tag{7}$$

Forward velocity:

$$v = \frac{0.5 + 0.3}{2} = 0.4\,\mathrm{m/s} \tag{8}$$

Angular velocity:

$$\omega = \frac{0.5 - 0.3}{0.20} = 1\,\mathrm{rad/s} \tag{9}$$

Radius of curvature (distance to ICR in diagram):

$$R = \frac{0.4}{1} = 0.4\,\mathrm{m} \tag{10}$$

This corresponds directly to the labeled R in the figure, indicating the robot follows a circular arc centered at the ICR.

Inverse kinematics are obtained by solving the forward equations:

$$\omega_r = \frac{2v + \omega b}{2r}, \quad \omega_l = \frac{2v - \omega b}{2r} \tag{11}$$

These equations define the mapping from desired motion (v, ω) to wheel actuation. However, the validity of this mapping depends on the assumptions illustrated in the diagram—namely, perfect rolling and no slip. In practical systems, deviations from these assumptions shift the effective ICR and introduce trajectory errors, forming the primary source of discrepancy between theoretical and observed motion.

4 Simulation and Error Analysis

To evaluate the impact of mechanical backlash on system behavior, a numerical simulation was conducted comparing ideal kinematic motion with a backlash-affected model. The simulation was implemented in discrete time using a forward Euler integration scheme, ensuring direct correspondence between the mathematical formulation and computational execution.

4.1 Simulation Model

Constant wheel angular velocities were applied:

$$\omega_r = 10 \text{ rad/s}, \quad \omega_l = 8 \text{ rad/s}$$

These inputs were selected to produce a smooth curved trajectory under ideal kinematic conditions.

Backlash in the wheel–shaft interface was modeled as a nonlinear transformation applied directly to the wheel angular velocities. For each wheel input ω, the effective velocity was computed as:

$$\omega_{\text{eff}} = \begin{cases} 0, & |\omega| \leq \theta_b \\ \omega - \text{sgn}(\omega)\theta_b, & |\omega| > \theta_b \end{cases}$$

where $\theta_b = 15^\circ = 0.2618$ rad represents the measured angular slack in the mechanical interface.

Given that the commanded wheel velocities satisfy $|\omega_r|, |\omega_l| \gg \theta_b$, the backlash model operates as a constant offset, reducing the effective angular velocity of each wheel. Thus:

$$\omega_{r,\text{eff}} = \omega_r - \theta_b, \quad \omega_{l,\text{eff}} = \omega_l - \theta_b$$

The linear and angular velocities for both models are then computed as:

$$v = \frac{r}{2}(\omega_r + \omega_l), \quad \omega = \frac{r}{b}(\omega_r - \omega_l)$$

for the ideal system, and

$$v = \frac{r}{2}(\omega_{r,\text{eff}} + \omega_{l,\text{eff}}), \quad \omega = \frac{r}{b}(\omega_{r,\text{eff}} - \omega_{l,\text{eff}})$$

for the backlash-affected system.

The robot pose is propagated in discrete time:

$$x_{k+1} = x_k + v_k \cos(\phi_k)\,\Delta t$$

$$y_{k+1} = y_k + v_k \sin(\phi_k)\,\Delta t$$

$$\phi_{k+1} = \phi_k + \omega_k\,\Delta t$$

where Δt is the simulation time step.

4.2 Trajectory Deviation Under Backlash

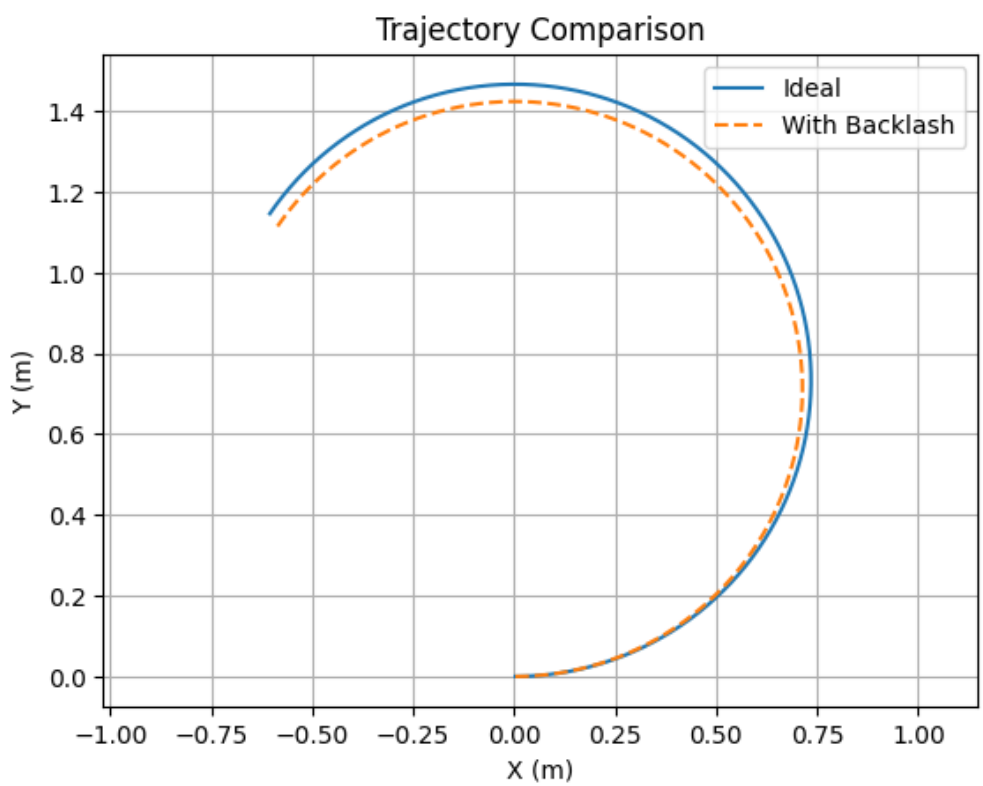

Figure 2: Comparison of ideal and backlash-affected trajectories under identical control inputs.

Figure 2 illustrates the resulting trajectories. The ideal model produces a circular arc consistent with the derived kinematic equations. In contrast, the backlash-affected trajectory exhibits a noticeable inward deviation and reduced curvature.

This deviation arises from the uniform reduction in wheel angular velocities. While the difference $(\omega_r - \omega_l)$ remains unchanged, preserving angular velocity ω, the average velocity $(\omega_r + \omega_l)$ decreases, reducing forward motion. Consequently, the effective turning radius

$$R = \frac{v}{\omega}$$

is reduced, causing the trajectory to contract inward relative to the ideal path.

4.3 Error Growth Over Time

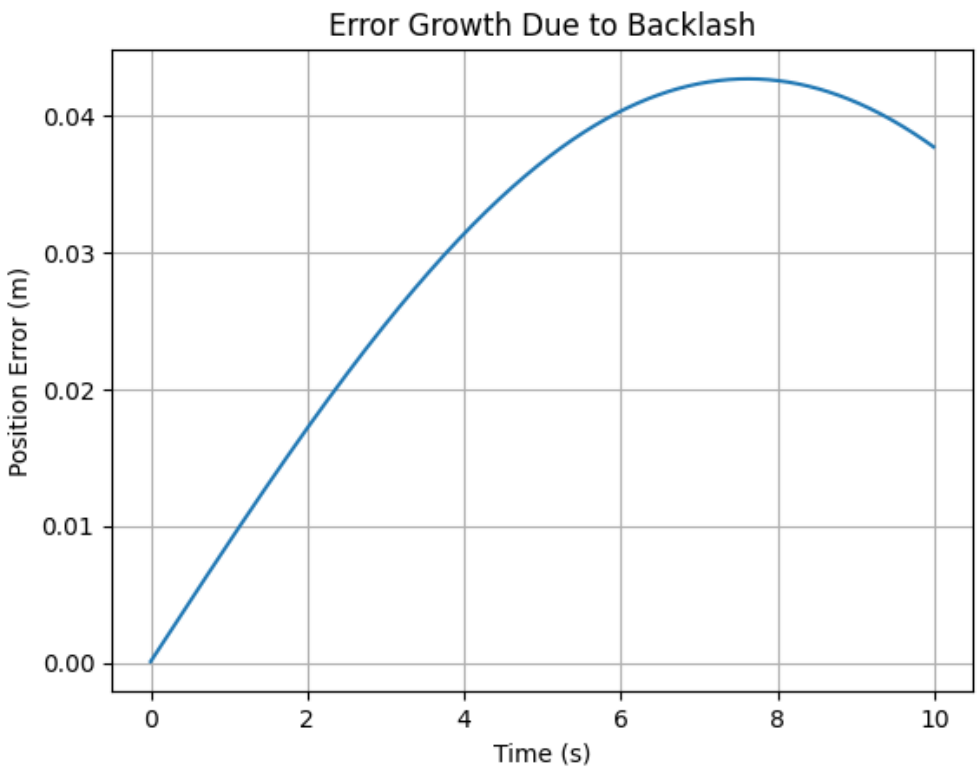

Figure 3: Growth of positional error between ideal and backlash-affected trajectories over time.

To quantify divergence, the positional error between the ideal and backlash trajectories is defined as:

$$e(t_k) = \sqrt{(x_i(t_k) - x_b(t_k))^2 + (y_i(t_k) - y_b(t_k))^2}$$

Figure 3 shows the evolution of this error over time. The error increases during the initial phase, reflecting cumulative divergence due to reduced forward velocity in the backlash model.

A gradual reduction in the rate of error growth is observed as the trajectories partially converge geometrically due to their similar angular velocities. However, the accumulated positional discrepancy remains non-negligible.

4.4 Interpretation

The simulation demonstrates that backlash, even when modeled as a simple constant offset in angular velocity, produces systematic and predictable distortion in robot motion. Unlike stochastic disturbances, this effect is deterministic and directly tied to mechanical design limitations.

Importantly, the angular velocity ω remains unchanged, while the linear velocity v is reduced. This decoupling leads to altered curvature and cumulative positional error, highlighting the sensitivity of open-loop kinematic systems to even minor actuation imperfections.

These results reinforce the central claim of this work: that ideal kinematic models are locally valid but fail to predict real-world behavior over extended operation in the absence of feedback.

5 System Architecture

The system is structured into sensing, computation, and actuation layers. The ESP32 functions as the central controller, receiving wireless input and generating motor commands. The L298N motor driver acts as the power interface, translating low-power control signals into high-current outputs for the motors.

The IMU is included but not used for feedback control. Its role is limited to post-hoc observation of orientation trends, ensuring that the system remains strictly open-loop during operation.

6 Implementation

Control inputs are mapped from joystick commands to desired linear and angular velocities. These values are converted into wheel angular velocities using inverse kinematics and then translated into PWM signals for motor actuation.

The implementation assumes a direct proportionality between PWM duty cycle and wheel angular velocity. This assumption is an approximation, as real motors exhibit nonlinear behavior influenced by load, voltage variation, and internal losses.

No calibration or system identification is performed, reinforcing the objective of observing uncorrected system behavior.

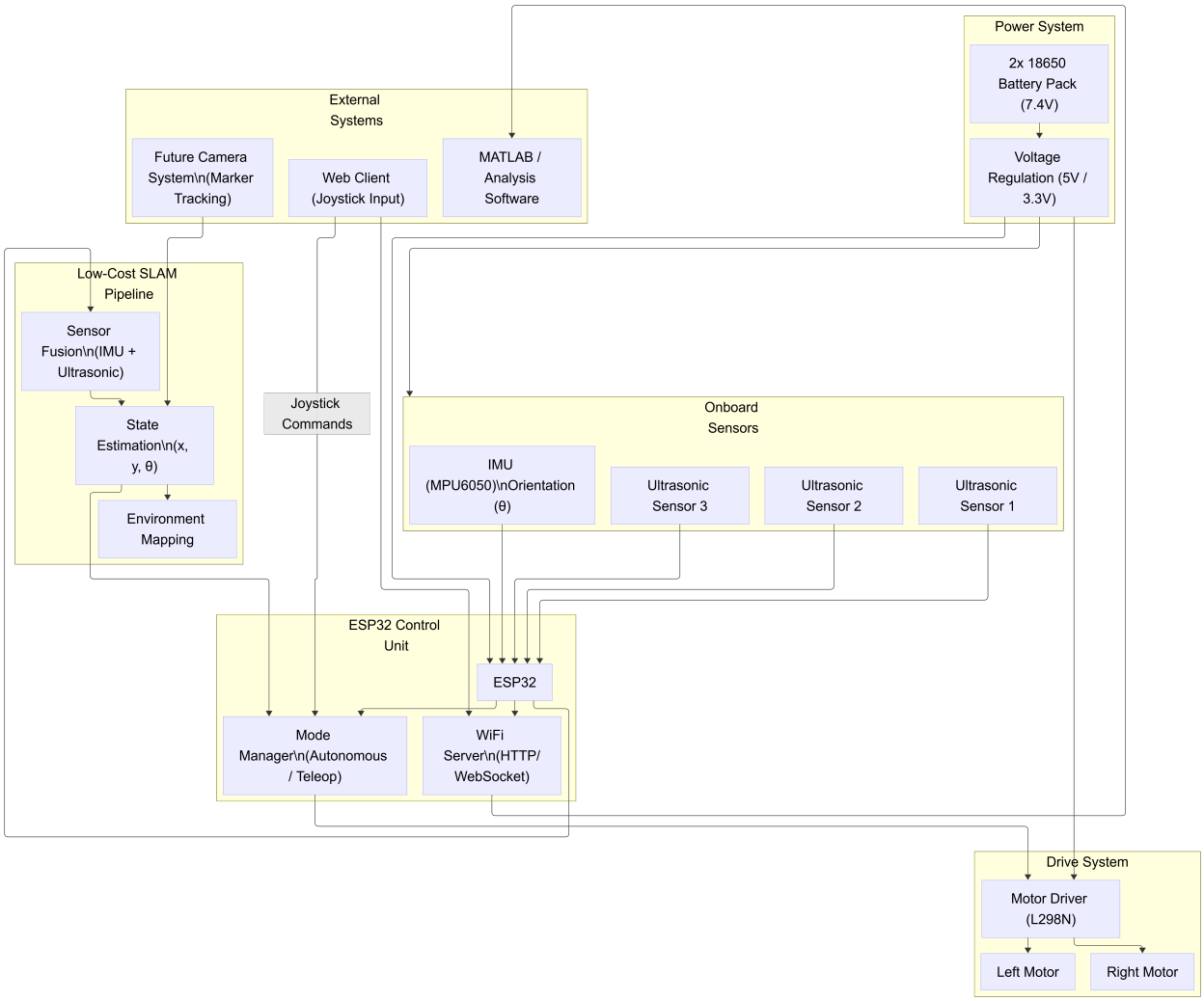

Figure 4: System Architecture

7 Results and Evaluation

Short-duration motion experiments show approximate agreement with theoretical predictions, indicating that the kinematic model is locally valid. However, deviations increase over time due to unmodeled dynamics such as wheel slip and actuator asymmetry.

Turning experiments confirm that curvature is governed by differential wheel velocity, but trajectories deviate from ideal circular paths. This deviation is primarily caused by unequal motor response and frictional inconsistencies.

Drift is consistently observed during straight-line motion, indicating that even small asymmetries produce cumulative error. Since no feedback correction is applied, these errors remain unbounded over time.

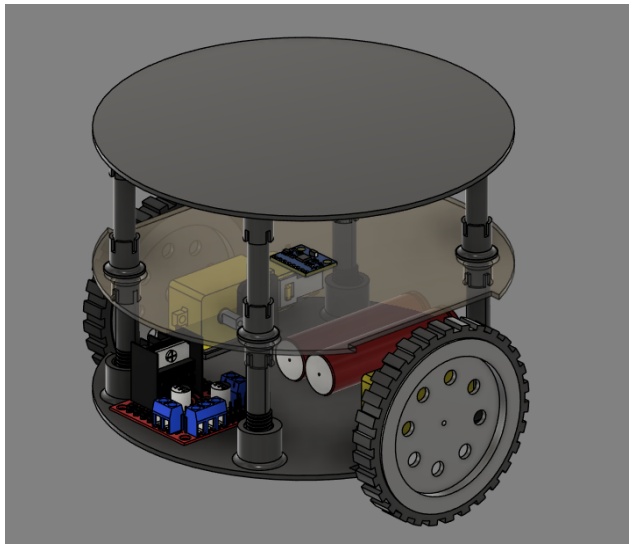

Figure 5: 3D Model of the Robot

8 Discussion

The results confirm that differential drive kinematics provide an accurate instantaneous model but fail as a predictive model over extended durations without feedback. The root cause lies in the violation of modeling assumptions, particularly pure rolling and actuator symmetry.

The system highlights a fundamental principle: kinematic correctness does not imply control reliability. Without state estimation or feedback, even a theoretically sound model cannot maintain accurate motion.

The trade-off between simplicity and performance is explicit. Removing sensing reduces system complexity but eliminates the ability to correct deviations. The robot therefore functions effectively as a demonstrator of open-loop limitations rather than a practical navigation system.

9 Conclusion

This work presents a differential drive robot designed to study the limitations of open-loop kinematic control. The theoretical model is derived and implemented successfully, and its validity is confirmed under short-duration conditions.

However, the absence of feedback leads to rapid degradation in positional accuracy, demonstrating the necessity of sensing and control in mobile robotics. The system serves as a foundational platform for future extensions involving state estimation and autonomous

behavior.

10 Future Work

Future development will focus on introducing feedback mechanisms to improve motion accuracy. Wheel encoders will enable odometry-based correction, while IMU integration can support orientation estimation.

Further extensions include visual sensing for localization and implementation of SLAM algorithms. Sensor fusion techniques will be explored to combine multiple data sources into a consistent state estimate.

These additions will transition the system from an open-loop demonstrator to a closed-loop autonomous platform.

11 Acknowledgement

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The development of this system also benefited from open-source documentation and prior work in mobile robotics, particularly in the domain of differential drive kinematics, which informed both the modeling and implementation stages.

Finally, the author acknowledges the iterative process of experimentation and failure that contributed significantly to the insights presented in this report.

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